

Case Study Part 1: Menopausal Hormone Therapy

Overview

- ▶ What is menopause?
- ▶ What is Menopausal Hormone Therapy?
- ▶ Status of research prior to Women's Health Initiative
- ▶ The Women's Health Initiative
- ▶ Risk as an estimand
- ▶ Recent observational studies & next steps

What is menopause?

- ▶ When menstruation pauses (stops)

What is menopause?

- ▶ When menstruation pauses (stops)
- ▶ Officially in menopause when you have gone 12 months without menstruating
 - ▶ The time leading up to menopause is called peri-menopause
 - ▶ After the 12-month mark, you are "menopausal" or postmenopausal

“Symptoms” in Menopause and Peri-menopause

- ▶ Referring to the experiences during (peri-)menopause as symptoms is in itself controversial

“Symptoms” in Menopause and Peri-menopause

- ▶ Referring to the experiences during (peri-)menopause as symptoms is in itself controversial
 - ▶ Because menopause is not a “disease”

“Symptoms” in Menopause and Peri-menopause

- ▶ Referring to the experiences during (peri-)menopause as symptoms is in itself controversial
 - ▶ Because menopause is not a “disease”
 - ▶ But we will use “symptoms” to describe the physical and mental health experiences attributable to menopause

"Symptoms" in Menopause and Peri-menopause

- ▶ Referring to the experiences during (peri-)menopause as symptoms is in itself controversial

- ▶ Because menopause is not a "disease"

- ▶ But we will use "symptoms" to describe the physical and mental health experiences attributable to menopause

- ▶ diminished sleep quality

- ▶ insomnia

- ▶ irritability

- ▶ depression

- ▶ mood swings

- ▶ difficulty concentrating

- ▶ reduced quality of life

- ▶ aching joints

- ▶ bloating

- ▶ breast tenderness

- ▶ bladder/urinary discomfort

- ▶ more frequent urinary tract infections

- ▶ vaginal dryness, discharge, and bleeding

- ▶ loss of sexual desire, painful intercourse

- ▶ headaches

- ▶ thinning hair

- ▶ heart palpitations

Menopausal Hormone Therapy: History

- ▶ What is (o)estrogen?

Menopausal Hormone Therapy: History

- ▶ What is (o)estrogen?
 - ▶ Everyone has hormones! Everyone has estrogen(s)– different types and different quantities

Menopausal Hormone Therapy: History

- ▶ What is (o)estrogen?
 - ▶ Everyone has hormones! Everyone has estrogen(s)– different types and different quantities
 - ▶ In females, primarily produced in ovaries and placenta (when pregnant); responsible for development and regulation of female reproductive system and secondary sex characteristics

Menopausal Hormone Therapy: History

- ▶ What is (o)estrogen?
 - ▶ Everyone has hormones! Everyone has estrogen(s)– different types and different quantities
 - ▶ In females, primarily produced in ovaries and placenta (when pregnant); responsible for development and regulation of female reproductive system and secondary sex characteristics
- ▶ 1940s: new techniques to extract estrogen from the urine of pregnant mares (female horses); marketed as a treatment for menopausal symptoms in the 1950s

Menopausal Hormone Therapy: History

- ▶ What is (o)estrogen?
 - ▶ Everyone has hormones! Everyone has estrogen(s)– different types and different quantities
 - ▶ In females, primarily produced in ovaries and placenta (when pregnant); responsible for development and regulation of female reproductive system and secondary sex characteristics
- ▶ 1940s: new techniques to extract estrogen from the urine of pregnant mares (female horses); marketed as a treatment for menopausal symptoms in the 1950s
- ▶ 1960s: "Feminine Forever": youth, beauty, full sex life for menopausal women who take estrogen

Menopausal Hormone Therapy: History

- ▶ What is (o)estrogen?
 - ▶ Everyone has hormones! Everyone has estrogen(s)– different types and different quantities
 - ▶ In females, primarily produced in ovaries and placenta (when pregnant); responsible for development and regulation of female reproductive system and secondary sex characteristics
- ▶ 1940s: new techniques to extract estrogen from the urine of pregnant mares (female horses); marketed as a treatment for menopausal symptoms in the 1950s
- ▶ 1960s: "Feminine Forever": youth, beauty, full sex life for menopausal women who take estrogen
- ▶ 1970s: Criticism of "paternalistic" medical establishment labeling menopause as a "deficiency disease" when it is a natural part of aging. Argue against using the term hormone "replacement"

Menopausal Hormone Therapy: History

- ▶ 1980s: Progesterone added for women with uterus as estrogen alone increased rates of uterine cancer and endometrial cancer (estrogen alone still used for those without uterus)

Menopausal Hormone Therapy: History

- ▶ 1980s: Progesterone added for women with uterus as estrogen alone increased rates of uterine cancer and endometrial cancer (estrogen alone still used for those without uterus)
- ▶ 1980s-1990s: multiple **observational** studies suggest hormones (estrogen or estrogen+progesterone) reduce risk of heart disease, reduce development of osteoporosis, might reduce the risk of Alzheimer's, and are the best available treatment for vasomotor symptoms

Menopausal Hormone Therapy: History

- ▶ 1980s: Progesterone added for women with uterus as estrogen alone increased rates of uterine cancer and endometrial cancer (estrogen alone still used for those without uterus)
- ▶ 1980s-1990s: multiple **observational** studies suggest hormones (estrogen or estrogen+progesterone) reduce risk of heart disease, reduce development of osteoporosis, might reduce the risk of Alzheimer's, and are the best available treatment for vasomotor symptoms
 - ▶ "...estrogen should be in the water... we thought it was like fluoride" -Hadine Joffe, Harvard Medical School professor of psychiatry

Observational Studies

- ▶ 1984 National Institutes of Health Consensus Conference on Osteoporosis statement: estrogens most effective means of preventing bone loss

Observational Studies

- ▶ 1984 National Institutes of Health Consensus Conference on Osteoporosis statement: estrogens most effective means of preventing bone loss
- ▶ Nurses' Health Study: 1991 observational study of 48,000 postmenopausal women: those who took hormones had a 50% lower risk of heart disease than those who did not

Observational Studies

- ▶ 1984 National Institutes of Health Consensus Conference on Osteoporosis statement: estrogens most effective means of preventing bone loss
- ▶ Nurses' Health Study: 1991 observational study of 48,000 postmenopausal women: those who took hormones had a 50% lower risk of heart disease than those who did not
- ▶ 1996 New York aging study: 1124 elderly women free of Alzheimer's, Parkinson's, and stroke; age at onset of Alzheimer's significantly later for those taking estrogen and relative risk significantly reduced compared to non-users

Observational Studies

- ▶ 1984 National Institutes of Health Consensus Conference on Osteoporosis statement: estrogens most effective means of preventing bone loss
- ▶ Nurses' Health Study: 1991 observational study of 48,000 postmenopausal women: those who took hormones had a 50% lower risk of heart disease than those who did not
- ▶ 1996 New York aging study: 1124 elderly women free of Alzheimer's, Parkinson's, and stroke; age at onset of Alzheimer's significantly later for those taking estrogen and relative risk significantly reduced compared to non-users
- ▶ 1998 study of Swedish women aged 50-81 with and without hip fractures; hormone therapy significantly reduces risk of hip fracture, risk of hip fracture decreased by about 6% for every year of use

Observational Studies

- ▶ 1984 National Institutes of Health Consensus Conference on Osteoporosis statement: estrogens most effective means of preventing bone loss
- ▶ Nurses' Health Study: 1991 observational study of 48,000 postmenopausal women: those who took hormones had a 50% lower risk of heart disease than those who did not
- ▶ 1996 New York aging study: 1124 elderly women free of Alzheimer's, Parkinson's, and stroke; age at onset of Alzheimer's significantly later for those taking estrogen and relative risk significantly reduced compared to non-users
- ▶ 1998 study of Swedish women aged 50-81 with and without hip fractures; hormone therapy significantly reduces risk of hip fracture, risk of hip fracture decreased by about 6% for every year of use
- ▶ Between 1975 and 2000, of twenty published studies examining menopausal hormone therapy, 80% found no increased risk of breast cancer, 10% found an increased risk of breast cancer, 10% found a decreased risk of breast cancer (meta-analysis sponsored by Wyeth)

The Women's Health Initiative

- ▶ under the National Institutes of Health (NIH)
- ▶ Launched in 1993
- ▶ Aims: test the benefits and risks of hormonal therapies (estrogen or estrogen+progesterone) for chronic diseases

The Women's Health Initiative

- ▶ under the National Institutes of Health (NIH)
- ▶ Launched in 1993
- ▶ Aims: test the benefits and risks of hormonal therapies (estrogen or estrogen+progesterone) for chronic diseases
- ▶ 16608 women without hysterectomy (intact uterus)
randomized: 8506 assigned to estrogen+progesterone (CEE+MPA), 8102 assigned to placebo

The Women's Health Initiative

- ▶ under the National Institutes of Health (NIH)
- ▶ Launched in 1993
- ▶ Aims: test the benefits and risks of hormonal therapies (estrogen or estrogen+progesterone) for chronic diseases
- ▶ 16608 women without hysterectomy (intact uterus)
randomized: 8506 assigned to estrogen+progesterone (CEE+MPA), 8102 assigned to placebo
- ▶ Intervention phase: November 1993-July 2002

The Women's Health Initiative

- ▶ under the National Institutes of Health (NIH)
- ▶ Launched in 1993
- ▶ Aims: test the benefits and risks of hormonal therapies (estrogen or estrogen+progesterone) for chronic diseases
- ▶ 16608 women without hysterectomy (intact uterus)
randomized: 8506 assigned to estrogen+progesterone (CEE+MPA), 8102 assigned to placebo
- ▶ Intervention phase: November 1993-July 2002
- ▶ Study halted July 2002 (average follow-up 5.2 years)

WHI Halted

- ▶ Why did the WHI halt the estrogen+progesterone arm of the study in 2002?

WHI Halted

- ▶ Why did the WHI halt the estrogen+progesterone arm of the study in 2002?
- ▶ Increased risks of breast cancer, stroke, heart attack, and clots (pulmonary embolism)

WHI Halted

- ▶ Why did the WHI halt the estrogen+progesterone arm of the study in 2002?
- ▶ Increased risks of breast cancer, stroke, heart attack, and clots (pulmonary embolism)
- ▶ What do we mean by increased risks?

WHI Halted

- ▶ Why did the WHI halt the estrogen+progesterone arm of the study in 2002?
- ▶ Increased risks of breast cancer, stroke, heart attack, and clots (pulmonary embolism)
- ▶ What do we mean by increased risks?
 - ▶ In 1 year for 10,000 postmenopausal women taking estrogen+progesterone, 8 more will have invasive breast cancer, 7 more will have a heart attack, 8 more will have a stroke, 18 more will have blood clots, than a similar group of 10,000 women not taking the hormones (Dr. Rossouw, acting director WHI)

WHI Criticized

- ▶ The halt? Internal Validity? External Validity?

WHI Criticized

- ▶ The halt? Internal Validity? External Validity?
- ▶ Criticism of the halt: at the time (July 2002), differences in risks across treatment and control groups not statistically significant

WHI Criticized

- ▶ The halt? Internal Validity? External Validity?
- ▶ Criticism of the halt: at the time (July 2002), differences in risks across treatment and control groups not statistically significant
 - ▶ How many additional cases of breast cancer is too many?

WHI Criticized

- ▶ The halt? Internal Validity? External Validity?
- ▶ Criticism of the halt: at the time (July 2002), differences in risks across treatment and control groups not statistically significant
 - ▶ How many additional cases of breast cancer is too many?
- ▶ Internal validity:

WHI Criticized

- ▶ The halt? Internal Validity? External Validity?
- ▶ Criticism of the halt: at the time (July 2002), differences in risks across treatment and control groups not statistically significant
 - ▶ How many additional cases of breast cancer is too many?
- ▶ Internal validity:
 - ▶ No multiple outcomes adjustments, no multiple subgroup comparison adjustments

WHI Criticized

- ▶ The halt? Internal Validity? External Validity?
- ▶ Criticism of the halt: at the time (July 2002), differences in risks across treatment and control groups not statistically significant
 - ▶ How many additional cases of breast cancer is too many?
- ▶ Internal validity:
 - ▶ No multiple outcomes adjustments, no multiple subgroup comparison adjustments
- ▶ External validity:

WHI Criticized

- ▶ The halt? Internal Validity? External Validity?
- ▶ Criticism of the halt: at the time (July 2002), differences in risks across treatment and control groups not statistically significant
 - ▶ How many additional cases of breast cancer is too many?
- ▶ Internal validity:
 - ▶ No multiple outcomes adjustments, no multiple subgroup comparison adjustments
- ▶ External validity:
 - ▶ "healthy women"

WHI Criticized

- ▶ The halt? Internal Validity? External Validity?
- ▶ Criticism of the halt: at the time (July 2002), differences in risks across treatment and control groups not statistically significant
 - ▶ How many additional cases of breast cancer is too many?
- ▶ Internal validity:
 - ▶ No multiple outcomes adjustments, no multiple subgroup comparison adjustments
- ▶ External validity:
 - ▶ "healthy women"
 - ▶ application of results to peri-menopausal women

WHI Criticized

- ▶ The halt? Internal Validity? External Validity?
- ▶ Criticism of the halt: at the time (July 2002), differences in risks across treatment and control groups not statistically significant
 - ▶ How many additional cases of breast cancer is too many?
- ▶ Internal validity:
 - ▶ No multiple outcomes adjustments, no multiple subgroup comparison adjustments
- ▶ External validity:
 - ▶ "healthy women"
 - ▶ application of results to peri-menopausal women
 - ▶ contextualization of risks

WHI Results

CEE + MPA Trial				
No. (%) of Events ^a				
	CEE + MPA (n = 8506)	Placebo (n = 8102)	Difference/ 10 000 PY ^b	HR (95% CI)
Primary end points				
Coronary heart disease				
50-59 y	38 (0.23)	27 (0.17)	5	1.34 (0.82-2.19)
60-69 y	79 (0.37)	73 (0.37)	0	1.01 (0.73-1.39)
70-79 y	79 (0.82)	59 (0.63)	19	1.31 (0.93-1.84)
Invasive breast cancer				
50-59 y	55 (0.33)	42 (0.27)	6	1.21 (0.81-1.80)
60-69 y	97 (0.45)	75 (0.38)	8	1.20 (0.89-1.62)
70-79 y	54 (0.56)	38 (0.41)	15	1.37 (0.90-2.07)

WHI Results

Age	N CEE+MPA	CEE+MPA B.C.	Prop.	N Placebo	Placebo B.C.	Prop.	HR
50-59	2837	55	0.0194	2683	42	0.0157	1.21
60-69	3854	97	0.0252	3655	75	0.205	1.20
70-79	1815	54	0.0298	1764	38	0.0215	1.37
Pooled	8506	206	0.0242	8102	155	0.0191	1.24

Risk as an Estimand

- ▶ The study reports counts and hazard ratios

Risk as an Estimand

- ▶ The study reports counts and hazard ratios
 - ▶ Time is a factor (longitudinal): (a) women who did not have breast cancer at $t=1$ might develop it at $t=2$ or $t=5$, (b) any individual's repeated follow-ups are not independent, (c) the numerator at later follow-ups may be different as people die

Risk as an Estimand

- ▶ The study reports counts and hazard ratios
 - ▶ Time is a factor (longitudinal): (a) women who did not have breast cancer at $t=1$ might develop it at $t=2$ or $t=5$, (b) any individual's repeated follow-ups are not independent, (c) the numerator at later follow-ups may be different as people die
- ▶ Survival Analysis and Hazard Ratios
 - ▶ Relate time that passes before an event occurs (e.g. breast cancer, stroke, death) to covariates

Risk as an Estimand

- ▶ The study reports counts and hazard ratios
 - ▶ Time is a factor (longitudinal): (a) women who did not have breast cancer at $t=1$ might develop it at $t=2$ or $t=5$, (b) any individual's repeated follow-ups are not independent, (c) the numerator at later follow-ups may be different as people die
- ▶ Survival Analysis and Hazard Ratios
 - ▶ Relate time that passes before an event occurs (e.g. breast cancer, stroke, death) to covariates
 - ▶ Among individuals without breast cancer, what is the probability they develop breast cancer in the next year?

Risk as an Estimand

- ▶ The study reports counts and hazard ratios
 - ▶ Time is a factor (longitudinal): (a) women who did not have breast cancer at $t=1$ might develop it at $t=2$ or $t=5$, (b) any individual's repeated follow-ups are not independent, (c) the numerator at later follow-ups may be different as people die
- ▶ Survival Analysis and Hazard Ratios
 - ▶ Relate time that passes before an event occurs (e.g. breast cancer, stroke, death) to covariates
 - ▶ Among individuals without breast cancer, what is the probability they develop breast cancer in the next year? The paper then pools this over all years into a hazard ratio (HR)

Risk as an Estimand

- ▶ How do we interpret hazard ratios?

Risk as an Estimand

- ▶ How do we interpret hazard ratios?
 - ▶ What if $HR = 1$?

Risk as an Estimand

- ▶ How do we interpret hazard ratios?
 - ▶ What if $HR = 1$? no difference between the hazard of breast cancer for treatment and control

Risk as an Estimand

- ▶ How do we interpret hazard ratios?
 - ▶ What if $HR = 1$? no difference between the hazard of breast cancer for treatment and control
 - ▶ What if $HR < 1$?

Risk as an Estimand

- ▶ How do we interpret hazard ratios?
 - ▶ What if $HR = 1$? no difference between the hazard of breast cancer for treatment and control
 - ▶ What if $HR < 1$? hazard of breast cancer is lower for treated

Risk as an Estimand

- ▶ How do we interpret hazard ratios?
 - ▶ What if $HR = 1$? no difference between the hazard of breast cancer for treatment and control
 - ▶ What if $HR < 1$? hazard of breast cancer is lower for treated
 - ▶ What if $HR > 1$?

Risk as an Estimand

- ▶ How do we interpret hazard ratios?
 - ▶ What if $HR = 1$? no difference between the hazard of breast cancer for treatment and control
 - ▶ What if $HR < 1$? hazard of breast cancer is lower for treated
 - ▶ What if $HR > 1$? hazard of breast cancer is higher for treated

Risk as an Estimand

- ▶ How do we interpret hazard ratios?
 - ▶ What if $HR = 1$? no difference between the hazard of breast cancer for treatment and control
 - ▶ What if $HR < 1$? hazard of breast cancer is lower for treated
 - ▶ What if $HR > 1$? hazard of breast cancer is higher for treated
- ▶ In 2002, the WHI Data and Safety Monitoring Board calculated an HR of 1.26, a 26% increase in breast cancer risk under treatment

Risk as an Estimand

- ▶ How do we interpret hazard ratios?
 - ▶ What if $HR = 1$? no difference between the hazard of breast cancer for treatment and control
 - ▶ What if $HR < 1$? hazard of breast cancer is lower for treated
 - ▶ What if $HR > 1$? hazard of breast cancer is higher for treated
- ▶ In 2002, the WHI Data and Safety Monitoring Board calculated an HR of 1.26, a 26% increase in breast cancer risk under treatment
- ▶ The WHI and media reporting on the halt of the study have been criticized for failing to communicate the difference between absolute and relative risks ($\frac{8}{10000} \neq .26$)

Results from WHI

What is the effect of estrogen+progesterone on invasive breast cancer in post-menopausal women?

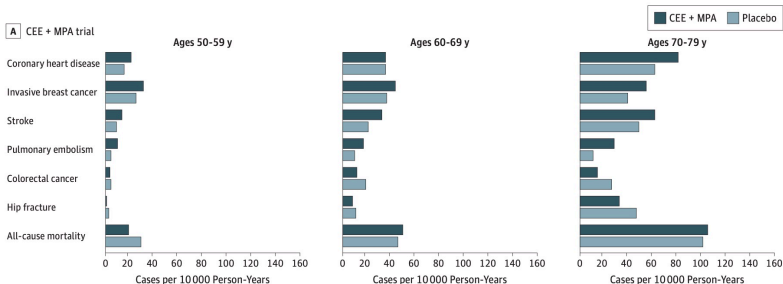
- ▶ Intervention phase (1993-2002): HR of 1.24 (95% CI, 1.01-1.53)

Results from WHI

What is the effect of estrogen+progesterone on invasive breast cancer in post-menopausal women?

- ▶ Intervention phase (1993-2002): HR of 1.24 (95% CI, 1.01-1.53)
- ▶ Overall, 9 more cases of invasive breast cancer per 10,000 person-years

Figure 3. Absolute Risks of Health Outcomes by 10-Year Age Groups in the Women's Health Initiative Hormone Therapy Trials During the Intervention Phase



WHIMS

- ▶ Prior observational studies suggested hormone therapy could reduce risk of Alzheimer's/cognitive decline or increase age of onset

WHIMS

- ▶ Prior observational studies suggested hormone therapy could reduce risk of Alzheimer's/cognitive decline or increase age of onset
- ▶ Who enrolled in WHIMS?

WHIMS

- ▶ Prior observational studies suggested hormone therapy could reduce risk of Alzheimer's/cognitive decline or increase age of onset
- ▶ Who enrolled in WHIMS? WHI 65 to 79 year olds, free of dementia at the time of enrollment

WHIMS

- ▶ Prior observational studies suggested hormone therapy could reduce risk of Alzheimer's/cognitive decline or increase age of onset
- ▶ Who enrolled in WHIMS? WHI 65 to 79 year olds, free of dementia at the time of enrollment
- ▶ CEE+MPA Arm: 2131 assigned to treatment, 2213 placebo; mean follow-up 4.2 years

WHIMS

- ▶ Prior observational studies suggested hormone therapy could reduce risk of Alzheimer's/cognitive decline or increase age of onset
- ▶ Who enrolled in WHIMS? WHI 65 to 79 year olds, free of dementia at the time of enrollment
- ▶ CEE+MPA Arm: 2131 assigned to treatment, 2213 placebo; mean follow-up 4.2 years
- ▶ What is the outcome of interest?

WHIMS

- ▶ Prior observational studies suggested hormone therapy could reduce risk of Alzheimer's/cognitive decline or increase age of onset
- ▶ Who enrolled in WHIMS? WHI 65 to 79 year olds, free of dementia at the time of enrollment
- ▶ CEE+MPA Arm: 2131 assigned to treatment, 2213 placebo; mean follow-up 4.2 years
- ▶ What is the outcome of interest? Annual measure of 3MSE (cognition test) scored 0 to 100

WHIMS

- ▶ Prior observational studies suggested hormone therapy could reduce risk of Alzheimer's/cognitive decline or increase age of onset
- ▶ Who enrolled in WHIMS? WHI 65 to 79 year olds, free of dementia at the time of enrollment
- ▶ CEE+MPA Arm: 2131 assigned to treatment, 2213 placebo; mean follow-up 4.2 years
- ▶ What is the outcome of interest? Annual measure of 3MSE (cognition test) scored 0 to 100
 - ▶ Temporal and spatial orientation, immediate and delayed recall, executive function, naming, verbal fluency, abstract reasoning, writing, copying

WHIMS

- ▶ Prior observational studies suggested hormone therapy could reduce risk of Alzheimer's/cognitive decline or increase age of onset
- ▶ Who enrolled in WHIMS? WHI 65 to 79 year olds, free of dementia at the time of enrollment
- ▶ CEE+MPA Arm: 2131 assigned to treatment, 2213 placebo; mean follow-up 4.2 years
- ▶ What is the outcome of interest? Annual measure of 3MSE (cognition test) scored 0 to 100
 - ▶ Temporal and spatial orientation, immediate and delayed recall, executive function, naming, verbal fluency, abstract reasoning, writing, copying
 - ▶ Women who scored below cutpoints were given additional testing for dementia (follow ups continued regardless of dementia status)

Results from WHIMS

- ▶ We still have longitudinal challenge

Results from WHIMS

- ▶ We still have longitudinal challenge
- ▶ Why is adherence a concern?

Results from WHIMS

- ▶ We still have longitudinal challenge
- ▶ Why is adherence a concern?
 - ▶ Non-compliance is always a concern to some extent, but non-compliance here may be directly related to cognitive decline!

Results from WHIMS

- ▶ We still have longitudinal challenge
- ▶ Why is adherence a concern?
 - ▶ Non-compliance is always a concern to some extent, but non-compliance here may be directly related to cognitive decline!
- ▶ CEE+MPA difference from placebo averaged 0.18 points (SE 0.10)

Results from WHIMS

- ▶ We still have longitudinal challenge
- ▶ Why is adherence a concern?
 - ▶ Non-compliance is always a concern to some extent, but non-compliance here may be directly related to cognitive decline!
- ▶ CEE+MPA difference from placebo averaged 0.18 points (SE 0.10)
- ▶ Difference is "too small to have relevance in clinical practice"

Results from WHIMS

Table 2. Mean Modified Mini-Mental State Examination Scores by Time From WHI Randomization and Treatment Assignment

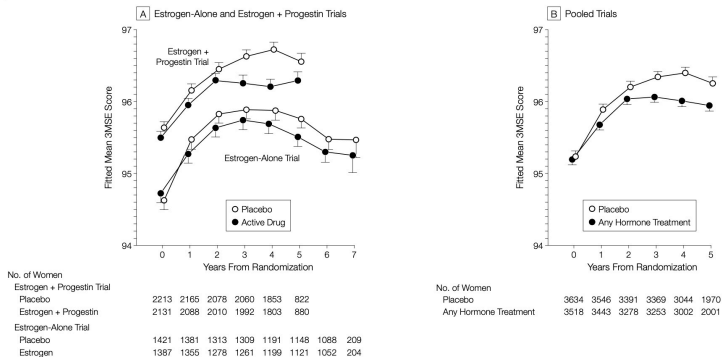
No. of Years Since Randomization	Treatment Assignment				Difference Between Treatments, Mean (95% CI)
	Estrogen + Progestin		Placebo		
	No. of Patients	Mean (SD)	No. of Patients	Mean (SD)	
0	2132	95.50 (4.21)	2215	95.63 (3.87)	−0.13 (−0.37 to 0.11)
1	2102	95.98 (4.10)	2188	96.20 (3.70)	−0.22 (−0.46 to 0.01)
2	2022	96.37 (3.92)	2095	96.50 (3.91)	−0.13 (−0.37 to 0.11)
3	2005	96.38 (4.36)	2083	96.74 (3.72)	−0.36 (−0.61 to −0.11)
4	1814	96.38 (4.96)	1875	96.90 (3.92)	−0.50 (−0.79 to −0.21)
5	833	96.71 (4.66)	901	96.87 (4.26)	−0.16 (−0.58 to 0.26)
6*	31	96.81 (2.39)	44	96.16 (8.63)	0.65 (−2.53 to 3.82)

Abbreviations: CI, confidence interval; WHI, Women's Health Initiative.

*Includes 9 women who were tested at 7 years after randomization.

Results from WHIMS

Figure 2. Fitted Mean Modified Mini-Mental State Examination Scores for Estrogen-Alone and Estrogen Plus Progestin Trials and Pooled Trials



WHI(MS) vs Observational Studies

Why are the results from the pre-WHI literature so different from the WHI(MS) results?

Recent Observational Studies

- ▶ Why consider evidence from observational studies?

Recent Observational Studies

- ▶ Why consider evidence from observational studies?
- ▶ Recent observational studies look at different formulations/doses, administrations, durations of use, such as:

Recent Observational Studies

- ▶ Why consider evidence from observational studies?
- ▶ Recent observational studies look at different formulations/doses, administrations, durations of use, such as:
 - ▶ micronized bioidential progesterones versus synthetic progesterone (MPA)
 - ▶ pill/tablets, patches, and injections
 - ▶ durations of use: 1 year or less, 1-3 years, 3 to 6 years, longer than 6 years

Next Steps

- ▶ Analyze data based on WHI and WHIMS for the effect of CEE+MPA on breast cancer and cognition score

Next Steps

- ▶ Analyze data based on WHI and WHIMS for the effect of CEE+MPA on breast cancer and cognition score
- ▶ Later: read about the effect of WHI/WHIMS on perceptions and prescriptions
- ▶ Later: analyze WHI- and WHIMS-based data with covariates
- ▶ Later: look closer at these recent observational studies with a focus on different levels of treatment